

RENA, Norway

WMO: 013890

Lat: 61.186N

Lon: 11.371E

Elev: 256

StdP: 98.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.2	(c) -23.2	(d) -28.2	(e) 0.3	(f) -26.2	(g) -25.1	(h) 0.4	(i) -23.3	(j) 8.2	(k) -0.5	(l) 7.1	(m) -1.3	(n) 0.7	(o) 130	(p) 0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	25.9	16.5	23.8	15.5	21.9	14.5	18.0	23.3	16.9	21.2	15.9	20.1	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.9	19.0	15.2	11.1	18.2	14.1	10.4	17.4	51.5	23.4	48.3	21.1	45.3	20.1	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.4	(b) 6.3	(c) 5.4	(e) -28.5	(f) 28.4	(g) 3.6	(h) 2.0	(i) -31.1	(j) 29.8	(k) -33.2	(l) 31.0	(m) -35.2	(n) 32.2	(o) -37.8	(p) 33.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-9.0	-7.7	-2.3	2.7	8.0	12.3	15.0	13.6	9.0	3.3	-2.6	-7.7
	DBStd	9.64	7.25	6.77	5.17	3.24	3.77	3.06	2.79	2.85	3.23	4.12	5.41	6.93
	HDD10.0	2977	590	496	380	218	84	11	4	55	211	378	547	
	HDD18.3	5623	848	729	638	468	320	181	110	149	279	467	628	806
	CDD10.0	402	0	0	0	1	23	81	155	115	26	2	0	0
	CDD18.3	8	0	0	0	0	0	1	6	1	0	0	0	0
	CDH23.3	212	0	0	0	0	20	47	112	32	0	0	0	0
	CDH26.7	31	0	0	0	0	2	8	20	1	0	0	0	0
Wind	WSAvg	2.0	1.8	2.0	2.1	2.1	2.3	2.4	2.1	1.9	1.9	1.9	1.8	1.8
Precipitation	PrecAvg	740	54	37	28	42	65	76	91	100	70	65	64	49
	PrecMax	988	107	93	83	97	149	152	213	194	141	160	204	90
	PrecMin	549	9	7	0	2	0	4	21	25	28	2	14	4
	PrecStd	107	26	19	20	25	37	32	39	38	29	38	41	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	6.2	11.5	17.8	26.0	28.1	29.2	26.2	21.1	14.9	10.0	6.9
		MCWB	3.9	3.0	5.6	10.4	14.5	16.6	18.9	16.4	13.8	11.4	8.6	6.0
	2%	DB	3.2	4.0	8.2	14.4	22.1	24.8	26.3	24.1	18.6	12.4	7.7	4.6
		MCWB	2.3	2.3	3.8	8.0	12.7	15.4	17.7	15.9	13.3	10.1	6.8	3.7
	5%	DB	1.5	2.4	6.2	11.8	19.5	22.1	24.2	22.1	16.5	11.0	5.8	2.9
		MCWB	0.7	1.2	2.6	6.7	11.7	13.8	16.2	15.3	12.5	9.4	5.1	2.0
	10%	DB	0.3	1.0	4.3	9.5	16.3	19.9	22.1	20.0	14.8	9.6	3.9	1.2
		MCWB	-0.1	0.1	1.7	5.3	10.0	12.9	15.1	14.7	11.8	8.2	3.4	0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.9	6.0	11.2	15.6	17.7	20.0	18.9	16.0	12.6	9.1	6.2
		MCDB	4.6	5.3	10.5	15.8	23.0	25.4	26.6	22.3	18.3	14.4	9.9	6.6
	2%	WB	2.4	2.5	4.3	8.8	13.8	16.3	18.6	17.5	14.4	10.8	6.8	3.9
		MCDB	3.0	3.6	7.4	13.0	20.1	22.4	24.5	21.1	16.9	11.8	7.4	4.4
	5%	WB	0.9	1.3	3.2	7.4	12.4	15.1	17.4	16.5	13.3	9.8	5.1	2.1
		MCDB	1.3	2.2	5.3	11.1	17.6	20.5	22.3	20.0	15.8	10.8	5.7	2.6
	10%	WB	-0.1	0.3	2.1	6.0	11.0	13.8	16.3	15.5	12.2	8.4	3.4	0.7
		MCDB	0.3	1.0	4.0	9.2	15.8	18.3	20.5	19.0	14.3	9.4	3.8	1.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	8.3	10.3	10.7	12.2	11.7	11.4	10.9	9.9	7.1	5.4	6.3
		MCDBR	6.1	8.1	12.3	15.7	18.9	17.0	16.2	15.2	13.3	9.1	6.7	7.1
		MCWBR	5.8	6.7	8.0	9.6	9.9	8.3	7.8	8.0	8.3	6.7	6.3	6.6
	5% WB	MCDBR	5.8	7.4	10.6	13.6	15.8	13.9	13.4	11.0	11.2	7.0	6.5	6.7
		MCWBR	5.5	6.2	7.4	8.7	8.7	7.3	7.0	6.6	7.4	5.7	6.1	6.2
Clear-Sky Solar Irradiance	taub	0.225	0.272	0.303	0.341	0.348	0.341	0.360	0.351	0.318	0.302	0.242	N/A	
	taud	2.146	2.260	2.268	2.303	2.430	2.481	2.442	2.461	2.535	2.485	2.231	N/A	
	Ebn at Noon	527	719	817	850	876	888	858	837	809	691	499	N/A	
	Edh at Noon	36	63	89	105	100	97	99	90	70	52	33	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.95	2.50	4.13	4.74	5.29	4.79	3.72	2.42	1.08	0.35	0.16	
	RadStd	0.05	0.19	0.23	0.43	0.46	0.48	0.61	0.33	0.29	0.19	0.08	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page